

Towards Enhancing Road Safety in South Carolina Using Insights from Traffic & Driver-Education Data

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Problem & Motivation

- SC has one of the highest traffic fatality rates in the U.S.
- \$2B+ yearly economic losses
- Increasing collision trends from 2017–2022
- Need for actionable insights and also stakeholder collaboration

Data

- Traffic collision data: 700,000+ records - (SC dept of Transportation)
- Alive@25 driver-education data
- 130MB of combined raw data
- U.S. Census county classifications (Urban / Mostly Rural / Rural)

Data Processing

Cleaning & Preparation

- 26.8% missing entries in traffic dataset
- 18.3% missing in Alive@25
- Removed invalid/out-of-state entries
- Standardized county-level comparison
- Built interactive dashboard

Dashboard

Interactive Tools Developed

- Time-based county choropleth map
- Collision heatmap
- Cumulative collision score
- User controls for year-by-year exploration

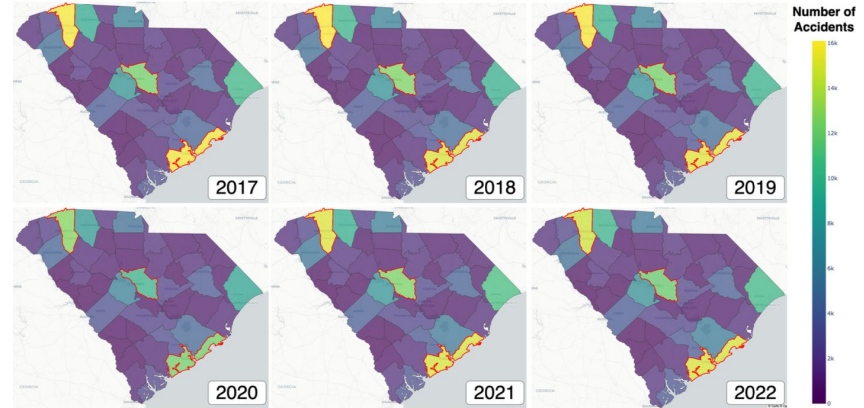


Figure 1: Heat map of South Carolina counties showing collision rates from 2017 to 2022. Lighter colors indicate higher collision rates, with Greenville, Charleston, and Richland counties highlighted as high-risk areas.

Key Findings

1. Geography-based patterns

- Urban counties (Greenville, Charleston, Richland) have highest collisions

2. Economic impact

- Rural counties have higher *average cost per incident*

3. Road-type issue

- Two-way undivided roads = highest crash frequency

4. Alive@25 program issue

- Participation decreasing in rural counties, collisions increasing

Example

Greenville County Case

- Largest collision volume 2017–2022
- Many crashes on two-way undivided roads
- Persistent hotspots visible on the dashboard

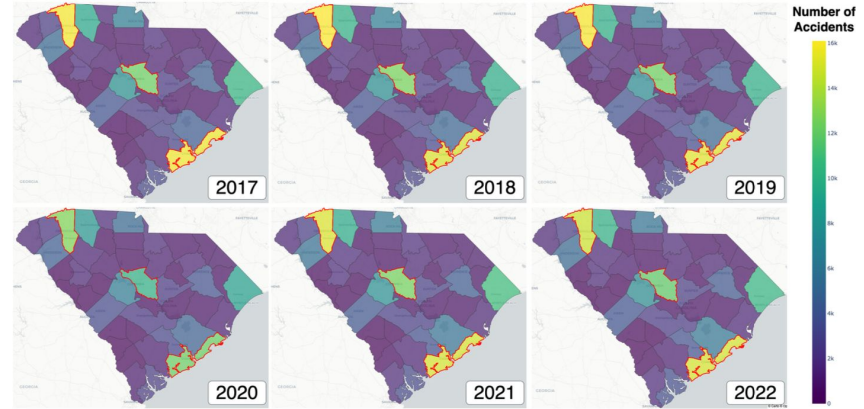


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Recommendations

Data-Driven Actions

1. Infrastructure

- Increase two-way divided roads with barriers in urban counties
- Prioritize Greenville, Charleston, Richland

2. Education

- Expand Alive@25 in mostly rural counties
- Address declining participation

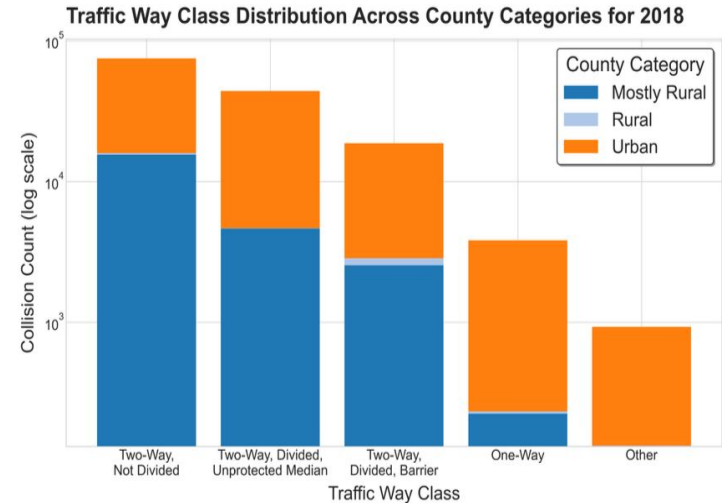


Figure 2: Stacked bar-plot highlighting the need for more two-way divided roads with barriers in urban counties to improve safety.

Critique

Pros & Cons

Pros

- Strong data cleaning
- Clear visualizations
- Multi-stakeholder collaboration

Cons

- No predictive modeling
- Dashboard not real time
- Could get more data like driver and weather which can impacts road accidents.

Conclusion & Future Work

Next Directions

- Adding predictive/simulation models
- Automate real-time dashboard updates
- Evaluate Alive@25 long-term effectiveness